Git Workflows

Advanced Embedded Software Development

with **Dan Walkes**



University of Colorado **Boulder**

Learning objectives:

Introduce Git Feature Branch Workflow

Introduce GitFlow

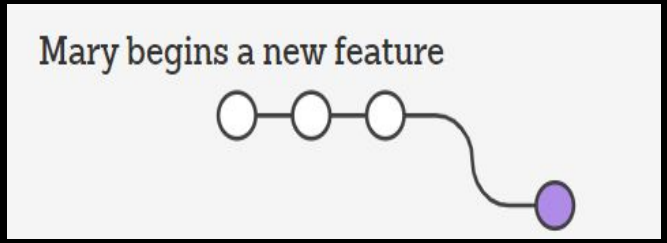
Git Feature Branch Workflow

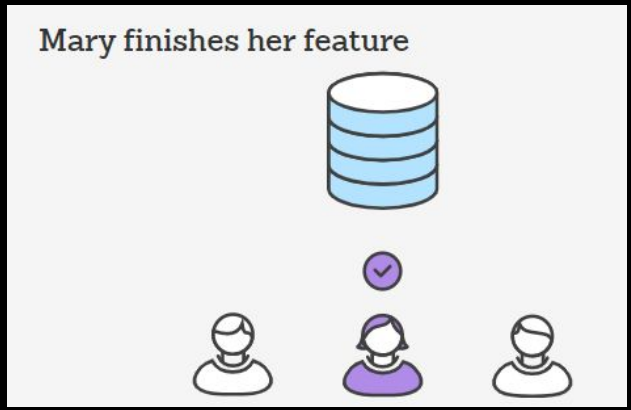
- Feature development takes place in a dedicated branch
  - Not in the master branch
- Multiple developers work on specific or different features without disturbing the main code base
- The master branch is never broken
- A feature ends with a pull request or merge request from a feature branch.
  - Code reviews happen on pull requests

https://www.atlassian.com/git/tutorials/comparing-workflows/feature-branch-workflow

Git Feature Branch Workflow

- git checkout master
- git pull origin master
- git checkout -b <feature branch name>
  - Often issue-25 - where issue 25 tracks the relevant github (or other) issue/bug.
  - Sometimes feature/foo
  - Sometimes fix-foo
- git commit ….
- git push origin <feature branch name>



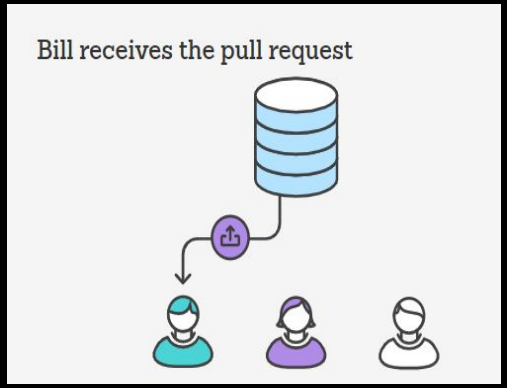


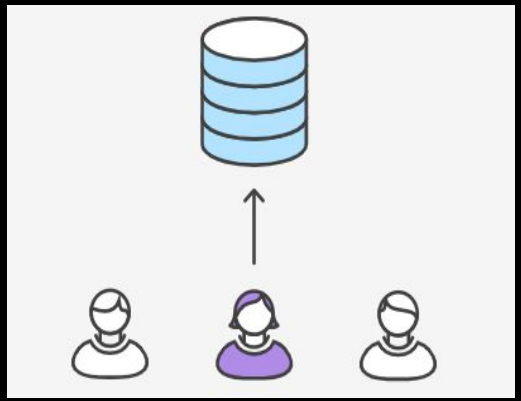
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Git Feature Branch Workflow

- Code review on <feature name branch>
  - Add comments, questions, suggestions

- Developer adds new changes based on pull request comments

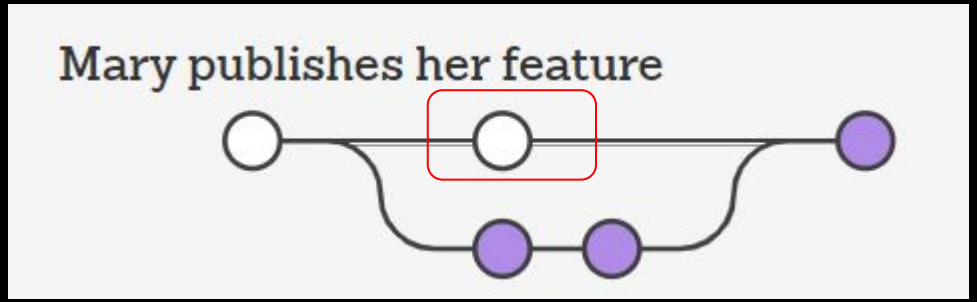




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Git Feature Branch Workflow

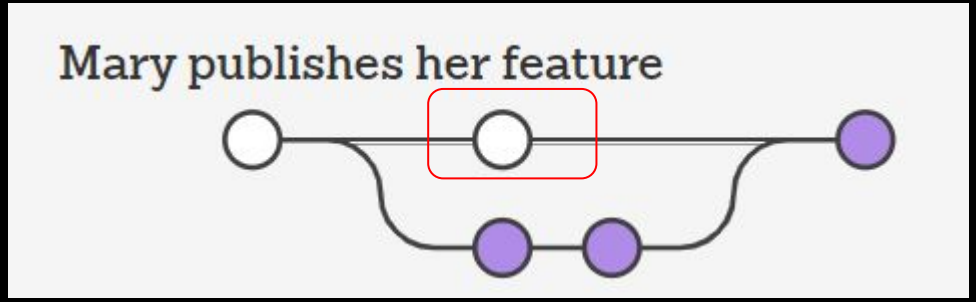
- Merge the feature into the master branch



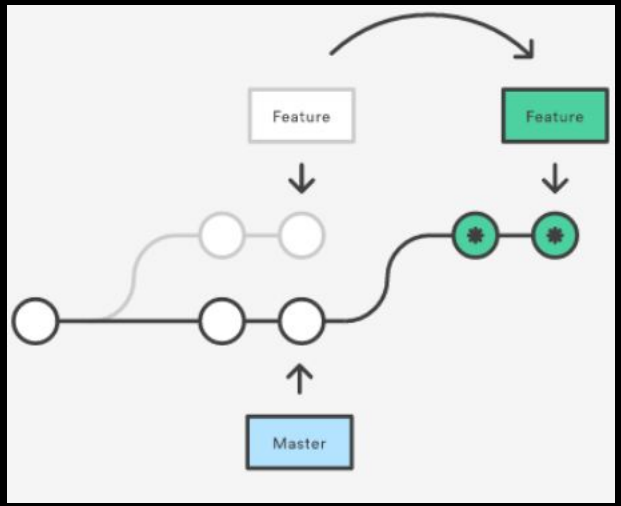
  - Why is this commit present?
    - Another feature was merged while this merge was in progress

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Git Feature Branch Workflow



- How can we handle this scenario?
  - Merge will add a merge commit to align the histories
  - Alternatively you can rebase the commit

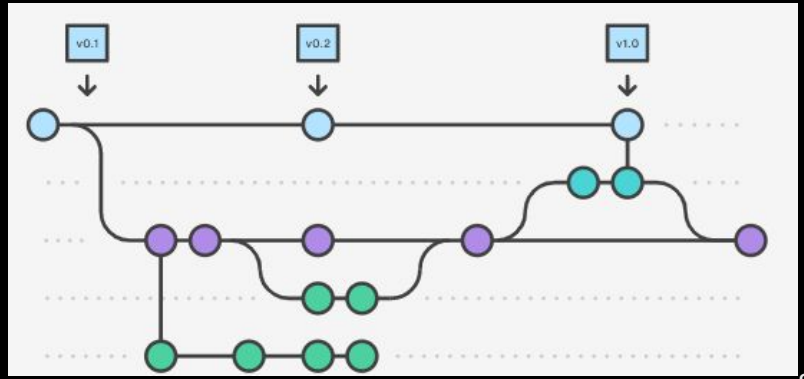


https://www.atlassian.com/git/tutorials/comparing-workflows/feature-branch-workflow
https://tekslate.com/git-rebase

GitFlow Workflow

- Based on Git Feature Branch Workflow
- Moves development onto a Develop branch
  - Master tracks release history
    - Only updated when building a release to customers/QA/etc
  - Develop branch is used for integration of features



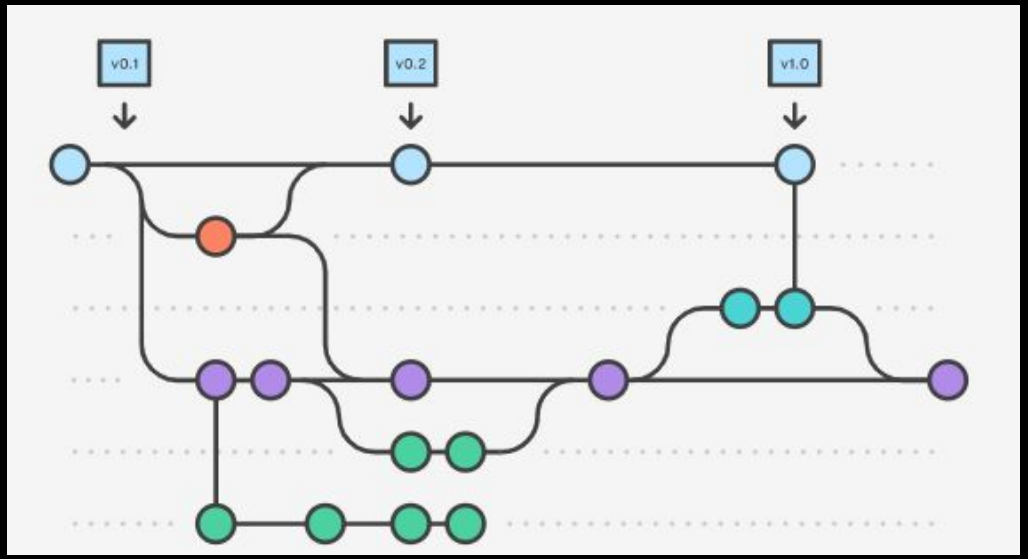


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GitFlow Workflow



- Hotfixes are based on master instead of develop



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